Echo Project Presentation Pack

Interview Flow

- 5-minute previous work presentation plus Q&A.
- Echocardiography segmentation / EF-estimation results presentation.
- Final Q&A on ML, DL, medical imaging, Python, English, and project expectations.

This PDF gives you

- Slide-ready baseline summary in English.
- Quantitative evidence for prediction collapse.
- Figures, tables, and talking points you can copy into slides.

Project Setup

Local dataset snapshot

- Training metadata rows: 400
- Available views: 2CH and 4CH sequences for train and test.
- Baseline experiment uses only 4CH videos.

Baseline pipeline

- Preprocess: load .npy sequence, keep first 10 frames, resize to 256x256, normalize per sequence.
- Split: 320 train / 80 validation, random seed 42.
- Model: TimeDistributed CNN + LSTM regressor with a final linear EF head.
- Outputs: saved weights, validation predictions, plots, and submission CSV.

Validation Summary

- The main story is variance collapse.
- Prediction mean/std = 51.18 / 0.0010
- Target mean/std = 53.41 / 11.01
- Negative R2 (-0.0418) means the baseline is worse than a mean predictor.

metric	value
Validation RMSE	11.1648
Validation MAE	9.0469
Validation R2	-0.0418
Prediction mean	51.1751
Prediction std	0.0010
Target mean	53.4100
Target std	11.0077
Prediction min	51.1700
Prediction max	51.1761
Target min	20.0000
Target max	77.7000
Median abs error	8.3522
Best-case abs error	0.0758
Worst-case abs error	31.1758
Manual RMSE check	11.1648
Manual R2 check	-0.0418

- Collapse evidence: prediction std is nearly zero while target std is large.
- This is strong numerical evidence that the model underfits patient-specific EF variation.

EF-Group Error Analysis

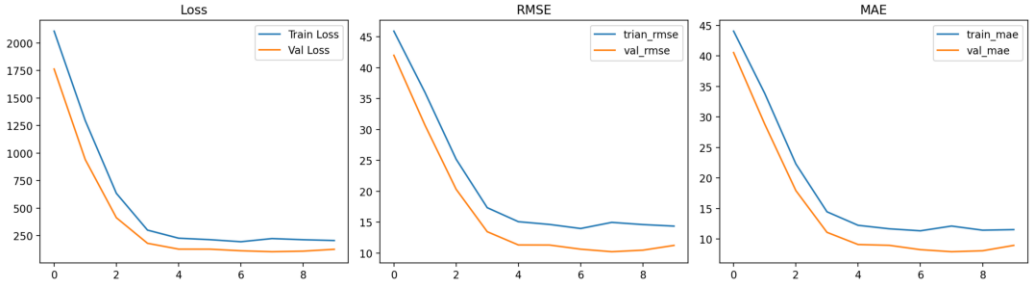
- Requested breakdown: EF < 40, 40 <= EF < 55, EF >= 55.
- Mean absolute error is lowest in the middle EF range and much higher at both extremes.

ef_group	n_samples	mean_target_ef	mean_prediction	mean_error	mean_abs_error	median_abs_error	max_abs_error
Low EF (<40)	9	33.4556	51.1754	17.7199	17.7199	15.9754	31.1758
Mid EF (40-55)	32	48.2656	51.1751	2.9095	4.1532	3.1246	10.6753
High EF (>=55)	39	62.2359	51.1750	-11.0609	11.0609	9.2240	26.5246

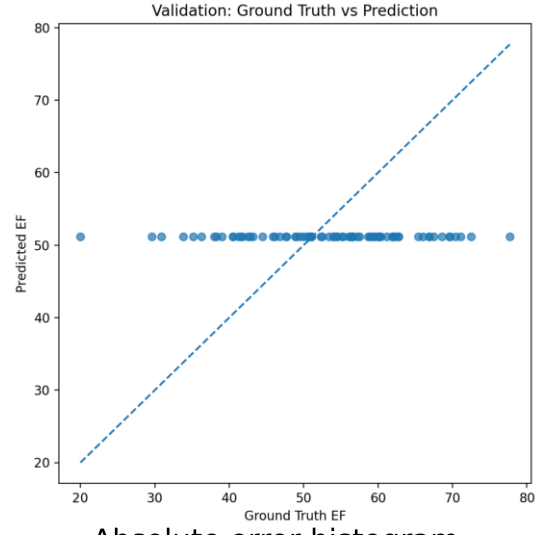
- Low EF mean abs error: 17.72
- Mid EF mean abs error: 4.15
- High EF mean abs error: 11.06

Core Result Figures

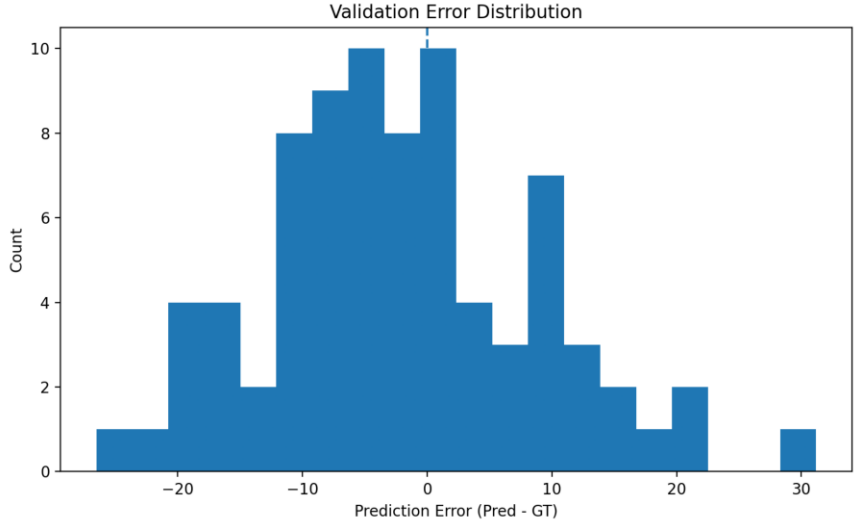
Training curves



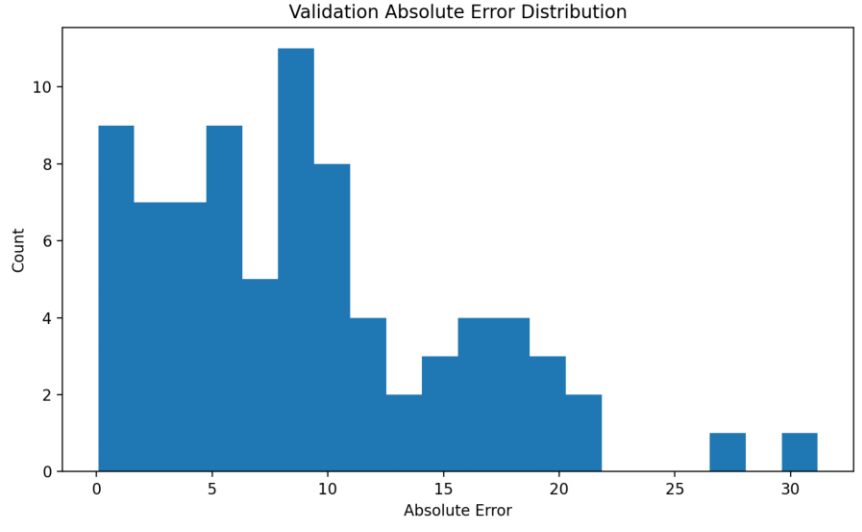
GT vs prediction scatter



Signed error histogram



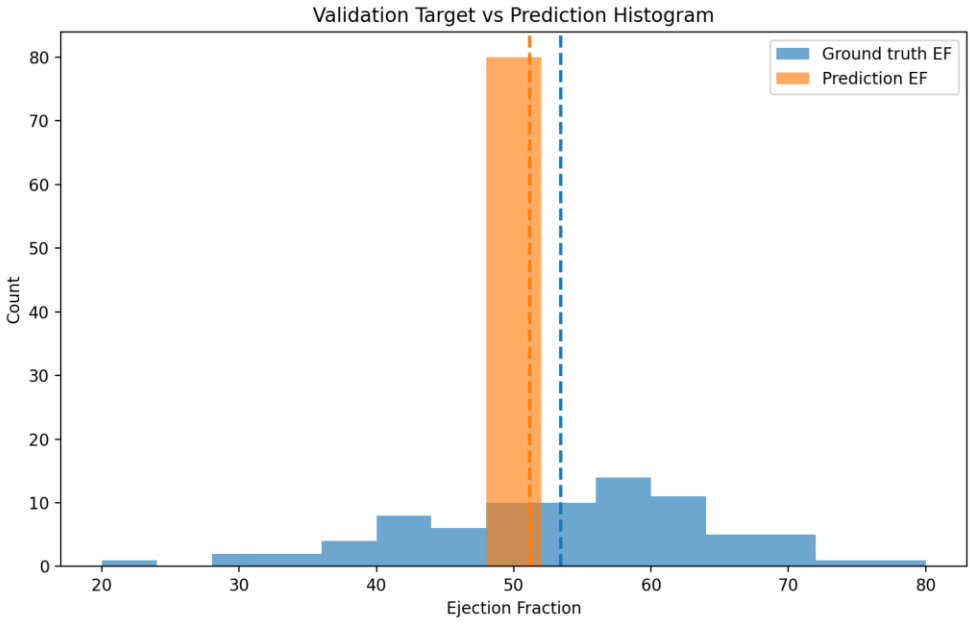
Absolute error histogram



- Scatter plot: weak alignment with the diagonal.
- Error histograms: the model makes both under- and over-estimates, but large tail errors remain.
- Training improves over epochs, but generalization remains poor.

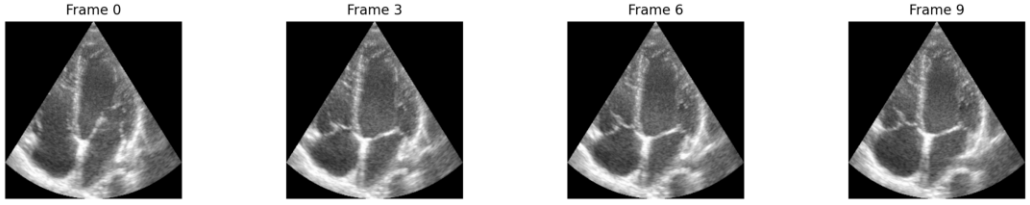
Prediction Collapse Evidence

GT vs prediction histogram



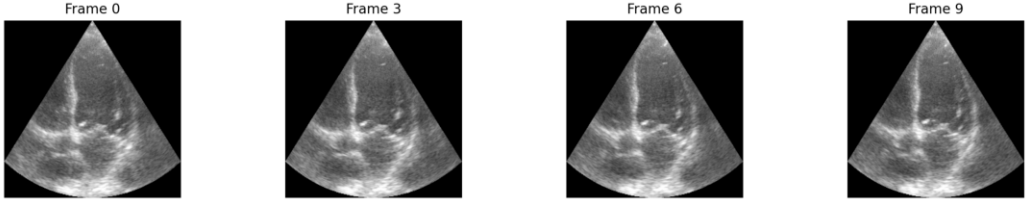
Representative best case

Best Case | patient174 | GT=51.10, Pred=51.18, AbsErr=0.08



Representative worst case

Worst Case | patient295 | GT=20.00, Pred=51.18, AbsErr=31.18



- Histogram overlap shows almost all predictions clustered around ~51 EF.
- Best cases are close because the target is already near the model's preferred prediction.
- Worst cases usually come from low-EF and high-EF extremes.

Slide-Ready Failure Analysis

Use this almost verbatim

- Validation RMSE: 11.16, MAE: 9.05, R2: -0.04.
- Predictions collapse near 51.18 EF.
- Scatter plot shows weak correlation with ground truth.
- Errors increase for low and high EF samples.
- Likely causes: limited temporal sampling, single-view input, simple architecture, no chamber-focused segmentation.

Professional framing

- This baseline is still useful because it gives a reproducible benchmark and a clean error diagnosis.
- The next model should focus on temporal modeling, view fusion, or segmentation-informed features.

Alternatives And Literature

EchoNet-Dynamic

- Official dataset site reports 10,030 labeled apical-4-chamber videos with measurements, tracings, and EF labels.
- Official paper reports direct EF prediction MAE around 7.0 and R2 around 0.50 on their benchmark.
- This is a strong optional reference for dataset size, code availability, and segmentation-aware supervision.

What to say in the interview

- The current repo establishes a regression baseline, but not a true LV segmentation pipeline.
- A stronger follow-up would segment the LV first, then estimate EF from EDV and ESV or from segmentation-guided video features.
- Main upgrade directions: 3D CNN / video transformer, two-view fusion, longer clips, explicit ventricle localization, and calibration-aware evaluation.